Sylvan Chen

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EDUCATION

University of Waterloo *Waterloo, Ontario*

Candidate for Bachelor of Computer Science (Honours) December, 2026

SKILLS

Systems & Backend: Python, C/C++, TypeScript, FastAPI, PostgreSQL, Redis, Kafka, React/Next.js

Data & AI Systems: LLM agent orchestration, retrieval pipelines (SQL/JSONB), eval & quality gating, streaming ingestion

Infrastructure: Docker, AWS, Railway, GitHub Actions, Jenkins, Git, Linux, CI/CD

PROFESSIONAL EXPERIENCE

AMP Energy Intelligence – *Founding Backend Engineer*

Toronto, ON Jun 2026 – Sep 2026

- As sole backend engineer, built a **0-to-1 multi-tenant analytics platform** with FastAPI, PostgreSQL, and AWS.
- Built **self-healing 5-minute/hourly ingestion** with idempotent writes, backfills, retries, and failure isolation.
- Redesigned rollups to cut p95 query latency from **9.3s to ⋆300ms** across **3.5M⋆** records at 32-way concurrency; **20×⋆** throughput.
- Secured **tenant isolation** with RBAC, encrypted credentials, scoped queries, and cross-tenant CI tests.

Linezero – *Software Engineer Intern*

Toronto, ON May 2026 – Sep 2026

- Built a **multi-tenant employee recognition platform** in React, FastAPI, and PostgreSQL, connecting peer recognition, award nominations, budget allocation, and reward redemption in one workflow.
- Integrated Workvivo APIs and webhooks to publish recognition updates and surface kudos-based prompts; added **OIDC SSO and SCIM 2.0 provisioning** with organization-scoped access for enterprise onboarding and deactivation.

CNEM Corporation – *Data & AI Engineering Intern*

Mississauga, ON Jan 2026 – Apr 2026

- Shipped a **Claude-powered research agent** with a two-pass quality gate for cross-entity retrieval, source verification, and evidence auditing, hitting **~80% Top-5 accuracy** on a 100+ query benchmark.
- Designed the **PostgreSQL knowledge layer** behind it: 5,000+ plant, compound, disease, mechanism, and PubChem entities in JSONB schemas built for cross-entity SQL retrieval.

Schneider Electric – *Software Engineer Intern*

Global R&D Jun 2025 – Sep 2025

- Built an **AI remediation pipeline for CI workflows** that fused static-analysis signals, dynamically retrieved code context, and an issue knowledge base into targeted pull-request fixes.
- Parallelized finding-level analysis with async Python agents, surfacing fixes inline via a VS Code CodeLens extension — **76% resolution rate** and **30%** faster PR-to-merge.

Eth Tech – *Software Engineer Intern*

Newark, CA Dec 2024 – Apr 2025

- Replaced batch ETL with **Kafka streaming consumers** for ML prediction services, sustaining **2,000⋆ events/sec** and halving processing lag through batching and retry/backoff.
- Added a Redis cache-aside layer for preference lookups, reaching a **~80% hit rate** and cutting downstream load in a high-availability production path.

PROJECTS

WatDFS Distributed File System – *C++ / FUSE / RPC* Jan 2026 – Apr 2026

- Implemented a **multi-client distributed filesystem** in C++ over FUSE: chunked RPC transfer, TTL/mtime cache validation, and write-back caching.
- Delivered **88.8× faster warm-cache reads** (110 vs. 1.24 MB/s) and ~1.9× scaling across two clients.